CHAPTER 14

BIODIVERSITY AND CONSERVATION

CHAPTER ESSENCE AT A GLANCE:
Analyze the three hierarchical levels of biodiversity, ecosystem services, global biodiversity hotspots, threats to biodiversity (HIPPO framework), IUCN Red List categories, and in situ versus ex situ conservation strategies.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Hierarchical Spectrum of Biodiversity:** Encompasses three interactive biological tiers: Genetic diversity (variation in genes within species), Species diversity (variety and abundance of species within a region), and Ecosystem diversity (variety of distinct biological habitats).

• **Vital Ecosystem Services:** Biodiversity performs essential ecosystem functions including nutrient cycling, organic matter decomposition, soil genesis, pollination of crops, water purification, climate regulation, and pharmaceutical discovery.

• **Global Biodiversity Hotspots:** Norman Myers identified regions featuring exceptional concentrations of endemic plant species that are undergoing rapid habitat destruction; criteria require at least 1500 endemic vascular plant species and loss of 70 percent of primary native vegetation.

• **Dual Conservation Methodologies:** In situ conservation protects endangered species within their native wild habitats (national parks, wildlife sanctuaries, biosphere reserves); Ex situ conservation preserves threatened genetic stock outside native environments (botanical gardens, gene banks, zoological parks).

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$H' = -\sum_{i=1}^{S} p_i \ln(p_i) \rightarrow \text{Shannon Wiener Species Diversity Index}$$

Information Theory Mathematical Measure of Species Richness and Evenness

$$S = cA^{z} \rightarrow \text{Species Area Power Law Function (z } \approx 0.25 \text{ to } 0.35)$$

Arrhenius Species-Area Power Scaling Relationship

$$\text{Biodiversity Loss (HIPPO)} = \text{Habitat Loss} + \text{Invasives} + \text{Pollution} + \text{Population} + \text{Overharvest}$$

Wilsonian Framework of Anthropogenic Extinction Drivers

$$\text{Hotspot Criteria} \rightarrow \geq 1500 \text{ Endemic Vascular Plants} \quad \text{and} \quad \geq 70\% \text{ Habitat Lost}$$

Norman Myers Strict Global Hotspot Qualification Criteria

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In a Biosphere Reserve, human settlements, tribal living, and commercial forestry are permitted within the outer Transition Zone, research is allowed in the Buffer Zone, but all human activity is strictly banned in the inner Core Zone.

• **Alert:** Habitat destruction and fragmentation is the single most catastrophic cause driving contemporary species extinction across the globe, far surpassing poaching and commercial hunting.

• **Alert:** Endemic species are plants and animals restricted strictly to a specific geographic territory and found nowhere else in the wild (e.g. the Lion-tailed Macaque in the Western Ghats of India).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define Biodiversity. Name its three hierarchical levels of organisation. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** The totality of genes, species, and ecosystems of a region, representing the variability among living organisms from all sources. **[1 Mark]**

• **Three Levels:** (1) Genetic Diversity, (2) Species Diversity, and (3) Ecosystem Diversity. **[1 Mark]**

2. What are Biodiversity Hotspots? State the two strict criteria required for an area to qualify as a hotspot. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** Biogeographical regions with exceptionally high levels of species richness and high endemism that face severe threats of habitat destruction. **[1 Mark]**

• **Two Criteria:** (1) Must contain at least 1500 species of vascular plants as endemics (more than 0.5 percent of the world's total), (2) Must have lost at least 70 percent of its original primary habitat. **[1 Mark]**

3. Distinguish between In situ and Ex situ conservation with two examples of each. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **In situ Conservation (On-site):** Protection of endangered flora and fauna within their natural wild habitats and evolutionary ecosystems (e.g. National Parks, Wildlife Sanctuaries, and Biosphere Reserves). **[1.5 Marks]**

• **Ex situ Conservation (Off-site):** Protection and maintenance of threatened animal and plant species outside their natural habitats in controlled artificial conditions (e.g. Botanical Gardens, Zoological Parks, and Seed and Cryo-gene Banks). **[1.5 Marks]**

4. Explain the ecological, economic, and scientific importance of biodiversity to human society. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Ecological Role:** Maintains ecosystem stability, regulates planetary hydrological and nutrient cycles, prevents soil erosion, and absorbs carbon dioxide. **[1 Mark]**

• **Economic Role:** Serves as the source of food, fiber, timber, industrial oils, and modern medicinal drugs derived from wild plants. **[1 Mark]**

• **Scientific and Ethical Value:** Each species holds irreplaceable genetic clues to evolutionary history and possesses an inherent right to exist. **[1 Mark]**

5. Discuss the major causes of biodiversity loss across the globe (the 'Evil Quartet' or HIPPO factors). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **1. Habitat Destruction and Fragmentation:** Clearing of rainforests for agriculture, logging, mining, and urbanisation destroys breeding grounds and fragments habitats. **[1 Mark]**

• **2. Over-exploitation:** Commercial over-hunting, poaching for ivory and horns, and industrial over-fishing deplete wildlife populations past recovery. **[1 Mark]**

• **3. Alien Invasive Species:** Introduction of non-native exotic species (e.g. Water Hyacinth, Lantana camara) outcompetes and drives native species extinct. **[1 Mark]**

• **4. Pollution and Climate Change:** Chemical pesticides, heavy metals, acid rain, and rising temperatures disrupt coral reefs and polar habitats. **[1 Mark]**

6. Describe the zoning structure of a Biosphere Reserve (Core, Buffer, and Transition Zones) and explain its multifaceted conservation objectives. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Core Zone:** Strictly protected, undisturbed ecosystem where human activity is completely prohibited, preserving wildlife in pristine wilderness. **[1.5 Marks]**

• **Buffer Zone:** Surrounds the core; permits controlled non-destructive activities like ecological research, education, and regulated ecotourism. **[1 Mark]**

• **Transition Zone:** Outermost boundary area where local communities, agriculture, sustainable forestry, and human settlements harmoniously coexist. **[1.5 Marks]**